Variance Reduction of Monte Carlo Regression Pricing of American Options

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DISCLAIMER

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1 Introduction

American options are widely traded in financial markets, yet their valuation remains a complex problem that has been the subject of extensive academic research. The least squares Monte Carlo approach proposed by Longstaff and Schwartz [**lsmc**] continues to stand out as one of the most competitive methods, particularly when it can be combined with variance reduction techniques [**chavez2022monte**].

For example, in the case of geometric Brownian motion processes, the use of control variates based on European options - following the framework proposed by Rasmussen [**rasmussen2005control**] yields significant improvements in accuracy, outperforming methods based on machine learning techniques such as multilayer neural networks (Chavez et al. [**chavez2022monte**]). However, this approach presupposes the ability to compute the exact price of a European option, which is typically feasible only in the case of simple price dynamics. More recently, Lin, Lin, and He [**lin2024analytically**] have demonstrated that it is possible to compute the prices of European options under stochastic volatility models, specifically under an improved Stein–Stein process (see section 2).

The objective of the present work is to extend the experiments originally conducted for geometric Brownian motion with constant volatility to the stochastic volatility setting developed in [**lin2024analytically**]. In particular, it seeks to validate the effectiveness of Rasmussen's control variates in this broader framework and to assess the relevance of considering alternative variance reduction techniques.

Write an overview of the report. Describe each section

2 The Model of the Price Dynamics

In section 3.3 we describe the algorithm for pricing an american put written on an underlying whose price process is assumed to follow the stochastic volatility model introduced by Lin, Lin, and He in [**lin2024analytically**]. This model is an improvement over the Stein-Stein stochastic volatility model introduced in [**stein1991stock**]. In [**lin2024analytically**] the authors, allow the long-run mean value of volatility to depend on time and to be stochastic. The price dynamics under the LLH model is given by the following system of SDEs:

$$\frac{dS_t}{S_t} = rdt + \sigma_t dW_t^1 \tag{1}$$

$$d\sigma_t = \kappa(\theta_t - \sigma_t)dt + \nu dW_t^2 \tag{2}$$

$$d\theta_t = \lambda dt + \eta dB_t \tag{3}$$

Where W_t^1 and W_t^2 are standard Brownian motions with correlation ρ, B_t is a standard geometric Brownian motion, and θ_t is the long-run mean value of volatility on a specific time interval t.

Remark 2.1

(i) $[W_t^1, W_t^2] = \operatorname{cov}(W_t^1, W_t^2) = \rho t$, where the brackets $[\cdot, \cdot]$ operator is that of quadratic covariation.

(ii) B_t is given by the following SDE:

$$dB_t = B_t dW_t, \quad B_0 = 1 \iff B_t = \exp\left(W_t - \frac{1}{2}t\right) \tag{4}$$

where W_t is a standard Brownian motion that is independent of W_t^1 and W_t^2.

Proposition 2.2 *For all $t \geq 0$,*

$$\theta_t = \theta_0 + \lambda t + \eta(B_t - B_0) \tag{5}$$

$$\begin{aligned}\sigma_t &= e^{-\kappa t}(\sigma_0 + \lambda - \theta_0) + \theta_0 + \lambda(t-1) + \nu W_t^2 - \\ &\qquad \nu\kappa \int_0^t e^{-\kappa s} W_{t-s}^2 ds + \eta\kappa \int_0^t e^{-\kappa s}(B_{t-s} - B_0) ds \end{aligned} \tag{6}$$

$$\begin{aligned}\textit{alternative}\ \sigma_t &= e^{-\kappa t}(\sigma_0 + \lambda - \theta_0) + \theta_0 + \lambda(t-1) \\ &\qquad + \eta(B_t - B_0) + e^{-\kappa t}\left(\eta \int_0^t e^{\kappa u} dB_u + \nu \int_0^t e^{\kappa u} dW_u^2\right)\end{aligned} \tag{7}$$

Proof

A direct integration of (3) yields

$$\theta_t = \theta_0 + \lambda t + \eta(B_t - B_0)$$

To integrate Eq. (2), proceed with the preliminary computation:

$$\begin{aligned} d(e^{\kappa t}\sigma_t) &= \kappa e^{\kappa t}\sigma_t dt + e^{\kappa t} d\sigma_t \\ &= \kappa e^{\kappa t}\sigma_t dt + \kappa e^{\kappa t}(\theta_t - \sigma_t) dt + \nu e^{\kappa t} dW_t^2 \\ &= \kappa e^{\kappa t}\theta_t dt + \nu e^{\kappa t} dW_t^2 \end{aligned}$$

Thus, direct integration yields σ_t in the following equation:

$$\begin{aligned} e^{\kappa t}\sigma_t - \sigma_0 &= \kappa \int_0^t e^{\kappa u}\left[\theta_0 + \lambda u + \eta(B_u - B_0)\right] du + \nu \int_0^t e^{\kappa u} dW_u^2 \\ &= \theta_0(e^{\kappa t} - 1) + \lambda\kappa \int_0^t u e^{\kappa u} du + \eta\kappa \int_0^t e^{\kappa u}(B_u - B_0) du + \nu \int_0^t e^{\kappa u} dW_u^2 \end{aligned}$$

Integration by parts simplifies the equation to

$$\begin{aligned} e^{\kappa t}\sigma_t - \sigma_0 &= \theta_0(e^{\kappa t} - 1) + \lambda\left[e^{\kappa t}(t-1) + 1\right] + \eta\kappa \int_0^t e^{\kappa u}(B_u - B_0)du \\ &\quad + \nu e^{\kappa t} W_t^2 - \nu\kappa \int_0^t e^{\kappa u} W_u^2 du \\ &= \lambda - \theta_0 + \theta_0 e^{\kappa t} + \lambda(t-1)e^{\kappa t} + \nu e^{\kappa t} W_t^2 - \\ &\quad \nu\kappa \int_0^t e^{\kappa u} W_u^2 du + \eta\kappa \int_0^t e^{\kappa u}(B_u - B_0)du. \end{aligned}$$

Thus,

$$\begin{aligned} \sigma_t &= e^{-\kappa t}(\sigma_0 + \lambda - \theta_0) + \theta_0 + \lambda(t-1) + \nu W_t^2 - \\ &\quad \nu\kappa \int_0^t e^{-\kappa(t-u)} W_u^2 du + \eta\kappa \int_0^t e^{-\kappa(t-u)}(B_u - B_0)du \end{aligned}$$

which yields (6) after a change of variable of $s = t - u$ in the integral terms □

Proposition 2.3 *For $t \geq 0$:*

$$S_t = S_0 \exp\left(rt - \frac{1}{2}\int_0^t \sigma_t^2 dt + \int_0^t \sigma_t dW_t^1\right) \tag{8}$$

amenable to bias-variance analysis

Proof Apply Ito's lemma to the log price process $X_t = \log S_t$ to obtain :

$$\begin{aligned} dX_t &= \frac{1}{S_t} dS_t - \frac{1}{2S_t^2} dS_t \cdot dS_t \\ &= \frac{dS_t}{S_t} - \frac{1}{2S_t^2}\sigma_t^2 S_t^2 dt \\ &= \left(r - \frac{1}{2}\sigma_t^2\right) dt + \sigma_t dW_t^1 \end{aligned}$$

Thus,

$$X_t = X_0 + rt - \frac{1}{2}\int_0^t \sigma_u^2 du + \int_0^t \sigma_u dW_u^1 \tag{9}$$

which yields (8) when X_t is inserted into $S_t = e^{X_t}$ $\square$

Remark 2.4

(i) Note that for $\eta = 0$:

$$\begin{aligned}\sigma_t &= e^{-\kappa t}(\sigma_0 + \lambda - \theta_0) + \theta_0 + \lambda(t-1) + \nu e^{-\kappa t}\int_0^t e^{\kappa u} dW_u^2 \\ &\sim \mathcal{N}\left(e^{-\kappa t}(\sigma_0 + \lambda - \theta_0) + \theta_0 + \lambda(t-1),\ \frac{\nu}{\kappa}\sinh \kappa t\right) \text{ revise}\end{aligned}$$

(ii) Notice that if the interval $[0, t]$ is discretized to $n+1$ points using a time step Δt, formula (6) yields the following estimator for σ_t:

$$\begin{aligned}\hat{\sigma}_t = e^{-\kappa t}(\sigma_0 + \lambda - \theta_0) + \theta_0 + \lambda(t-1) + \nu W_t^2 - \\ \nu\kappa\Delta t \sum_{i=1}^n e^{-\kappa i \Delta t} W_{t-i\Delta t}^2 + \eta\kappa\Delta t \sum_{i=1}^n e^{-\kappa i\Delta t}(B_{t-i\Delta t} - B_0)\end{aligned} \tag{10}$$

$$\begin{aligned}\text{alternative } \hat{\sigma}_t \approx e^{-\kappa t}(\sigma_0 + \lambda - \theta_0) + \theta_0 + \lambda(t-1) + \eta(B_t - B_0) \\ + e^{-\kappa t}\left(\eta \sum_{i=1}^n e^{\kappa(i-1)\Delta t}\Delta B_i + \nu \sum_{i=1}^j e^{\kappa(i-1)\Delta t}\Delta W_i^2\right)\end{aligned} \tag{11}$$

where ΔB_i and ΔW_i^2 are increments of the processes B_t and W_t^2 on the interval $[(i-1)\Delta t, i\Delta t]$, $i = 1, \ldots, n$ Notice the variance of $\hat{\sigma}_t$ may be computed, so may be useful in investigating convergence in MSE. Also, it may be worthwhile to review the estimates of the stochastic integrals.

(iii) Similarly, formulas (9) and (8) yield the following estimators of the price:

$$\hat{S}_t = S_0 e^{rt} \exp\left(-\frac{\Delta t}{2}\sum_{i=0}^{n-1}\sigma_i^2 + \sum_{i=0}^{n-1}\sigma_i \Delta W_i^1\right) \tag{12}$$

2.1 Simulating Paths from the Price Process

Given the parameters κ, ν, λ, η, and ρ, the price process path may be simulated by following the algorithm:

Algorithm 2.5

To simulate a price path up to time t, disrectize time to $n+1$ nodes $t_j = j\Delta t$ with time step Δt and $j = 0, \ldots, n$.

(i) Generate n i.i.d. samples Z_i from a standard normal. Set $\Delta W_i = \sqrt{\Delta t} Z_i$ and notice that

$$W_j := W_{t_j} = \sum_{i=1}^{j} \Delta W_i = \sqrt{\Delta t}\sum_{i=1}^{j} Z_i$$

then use Eq. (4) to compute B_t and its increments as follows:

$$B_j := B_{t_j} = e^{-\frac{j\Delta t}{2}} \exp W_j \tag{13}$$

$$\Delta B_j := B_j - B_{j-1} \tag{14}$$

for $j = 1, \ldots, n$.

(ii) Generate independently n i.i.d. samples Z_i^1 and Z_i^2 from a standard normal. Set the increments:

$$\Delta W_i^1 := \sqrt{\Delta t} Z_i^1 \text{ and } \Delta W_i^2 := \sqrt{\Delta t}\left(\rho Z_i^1 + \sqrt{1-\rho^2} Z_i^2\right) \text{ for } i = 1, \ldots, n$$

then compute

$$W_j^2 := W_{t_j}^2 = \sum_{i=1}^{j} \Delta W_i^2 = \sqrt{\Delta t} \sum_{i=1}^{j} \left(\rho Z_i^1 + \sqrt{1-\rho^2} Z_i^2 \right) \text{ for } j = 1, \ldots, n$$

(iii) For $j = 1, \ldots, n$ use Eqs. (10), (13), and (14) to estimate the volatilities:

$$\begin{aligned} \hat{\sigma}_j &:= \hat{\sigma}_{t_j} \\ &\approx e^{-\kappa j \Delta t}(\sigma_0 + \lambda - \theta_0) + \theta_0 + \lambda(j\Delta t - 1) + \nu W_j^2 - \\ &\quad \nu\kappa\Delta t \sum_{i=1}^{j} e^{-\kappa i \Delta t} W_{j-i}^2 + \eta\kappa\Delta t \sum_{i=1}^{j} e^{-\kappa i \Delta t}(B_{j-i} - 1), \end{aligned} \tag{15}$$

Notice that these estimates have been obtained as Riemann sum approximations of the definite integrals. One can well use more accurate and more efficient numerical integration methods (quadrature, guass-). Mind the exponents of Δt in the estimation of the volatility! Analyze the exponentially decaying terms as $\Delta t \to 0$.

$$\begin{aligned} \text{alternative}\hat{\sigma}_j &:= \hat{\sigma}_{t_j} \\ &\approx e^{-\kappa j \Delta t}(\sigma_0 + \lambda - \theta_0) + \theta_0 + \lambda(j\Delta t - 1) + \eta(B_j - 1) \\ &\quad + e^{-\kappa j \Delta t} \left(\eta \sum_{i=1}^{j} e^{\kappa(i-1)\Delta t} \Delta B_i + \nu \sum_{i=1}^{j} e^{\kappa(i-1)\Delta t} \Delta W_i^2 \right) \end{aligned}$$

(iv) For $j = 1, \ldots, n$ use Eq. (8) to estimate the price values:

$$\begin{aligned} \hat{S}_j &= \hat{S}_{t_j} \\ &= S_0 e^{rj\Delta t} \exp\left(-\frac{\Delta t}{2} \sum_{i=0}^{j-1} \sigma_i^2 + \sum_{i=0}^{j-1} \sigma_i \Delta W_i^1 \right) \end{aligned} \tag{16}$$

alternative Solve the following recurrence equation for $\hat{S}_j$, $j = 1, \ldots, n$:

$$\hat{S}_{i+1} - \hat{S}_i = \hat{S}_i \left(r\Delta t + \hat{\sigma}_i \Delta W_i^1 \right) \tag{17}$$

Notice that the recurrence relation yields:

$$\begin{aligned} \hat{S}_j &= S_0 \prod_{i=0}^{j-1} \left(1 + r\Delta t + \hat{\sigma}_i \Delta W_i^1 \right) \\ &= S_0 \prod_{i=0}^{j-1} \left(1 + r\Delta t + \hat{\sigma}_i \sqrt{\Delta t} Z_i^1 \right) \end{aligned} \tag{18}$$

Can I consider a more accurate discetization scheme of the price?

Remark 2.6 Notice that (13), (15), and (18) suggest the terms driving the evolution of the price (by eliminating the terms with higher powers in Δt!)

2.2 Estimation of the Model Parameters

Following [**hirsa2024computational**] and [**lin2024analytically**], we propose the following estimation method:

(i) Collect the market prices of European options for a grid of maturities τ_i and strike prices K_j.

(ii) Assuming that the model parameters were known, for given values of τ_i and K_j, an estimator of the price of the option $\widehat{V}_{ij}$ can be constructed using Monte Carlo simulations under the price model. (Is it faster to solve the dynamical system for the European price?)

(iii) Now the optimal parameters $\Theta = (\kappa, \nu, \lambda, \eta, \rho)$ may be found by solving the optimization problem:

$$\widehat{\Theta} = \text{argmin}_{\Theta} MSE(V_{ij}, \widehat{V}_{ij})$$

where V_{ij} is the market value of the option under the maturity τ_i and strike K_j.

Relate to the volatility smile

3 Pricing Options Under the LLH Model

We assume the risk free rate r to be fixed, otherwise r should be replaced by the discount factor $B(t,T) = \exp\left(-\int_t^T r_s ds\right)$. We denote by Q the martingale measure underlying the LLH model (1) - (3).

3.1 The European Price Formula

Under the dynamics (8), the price of a European call U with spot S_t, strike K, and maturity date T, is obtained as an expectation conditional on the information at time t:

$$\begin{aligned} U(t,T,r,K,S_t,\sigma_t,\theta_t) &= \mathbb{E}_t^Q[e^{-r(T-t)} \max(S(T)-K,0)] \\ &= \mathbb{E}_t^Q[e^{-r(T-t)}(S(T)-K)1_{\{S_T>K\}}] \\ &= \mathbb{E}_t^Q[e^{-r(T-t)}S(T)1_{\{S_T>K\}}] - Ke^{-r(T-t)}\mathbb{E}_t^Q[1_{\{S_T>K\}}] \end{aligned}$$

To simplify the computation of the first expectation, change the numeraire to $S(t)$ and the martingale measure to Q_1 whose Radon-Nikodym derivative given by

$$\frac{dQ_1}{dQ} = e^{-r(T-t)}\frac{S(T)}{S(t)} \tag{19}$$

Thus,

$$\begin{aligned}
U(t,T,r,K,S_t,\sigma_t,\theta_t) &= \mathbb{E}_t^{Q_1}[S(t)1_{\{S_T>K\}}] - Ke^{-r(T-t)}\mathbb{E}_t^{Q}[1_{\{S_T>K\}}] \\
&= S(t)Q_1[x_T > \ln K\,|S_t] - Ke^{-r(T-t)}Q[x_T > \ln K\,|S_t] \\
&= S(t)F^{Q_1}(\ln K) - Ke^{-r(T-t)}F^{Q}(\ln K) \\
&= S_tP_1 - Ke^{-r(T-t)}P_2
\end{aligned}$$

where $x_t = \ln S_t$, F^{Q_1} and F^Q are the distribution functions of x_T under Q_1 and Q respectively, conditional on the information available at time t. The diffusion equation of x_t is given in (9). Using the inversion theorem in [**gilPelaez**], the probabilities P_1 and P_2 may be recovered from the respective conditional characteristic functions $f_1(\phi)$ and $f_2(\phi)$ of x_T under Q_1 and Q. Note that,

$$f_1(\phi) = \mathbb{E}_t^{Q_1}[e^{i\phi x_T}] \quad \text{and} \quad f_2(\phi) = \mathbb{E}_t^{Q}[e^{i\phi x_T}]$$

and the inversion theorem yields

$$P_j = \tfrac{1}{2} + \frac{1}{\pi}\int_0^\infty Re\left[\frac{e^{-i\phi\ln K}}{i\phi}\,f_j\right]\,d\phi.$$

From here onward, we gleaned two approaches to compute the f_j's. One that is PDE based following Heston [**hestonPDE**], and the other is stochastic calculus based following Schobel and Zhu [**schobelZhu**]. In [**lin2024analytically**], the authors follow the former approach which using a standard martingale argument, a PDE for U is derived that ultimately yields

PDEs for the $P_j's$ and consequently the $f_j's$. Applying the ansatz from Lord and Kahl [**Lord2006Optimal**], for affine diffusion models the $f_j's$ must be exponential in the state variables, leads to the system of ODEs in [**lin2024analytically**].

We use the stochastic calculus machinery developed in [**schobelZhu**] to derive $f_1(\phi)$ in terms of the model variables and parameters. We use the following two facts in the derivation:

(i) Since $dW_t^1 dW_t^2 = \rho dt$, the following representation is obtained

$$dW_t^1 = \rho dW_t^2 + \sqrt{1-\rho^2} dW_t$$

where $dW dW_t^1 = \sqrt{1-\rho^2}$ and $dW dW_t^2 = 0$.

(ii)

$$\mathbb{E}_t^Q \left[\exp\left(\alpha \int_t^T H_u dW_u \right) \right] = \exp\left(\frac{1}{2}\alpha^2 \int_t^T H_u^2 du \right) \qquad (\alpha \in \mathbb{C}) \quad \text{(add ref)}$$

for as long as H is non-anticipative and independent of the Brownian motion W.

(iii) Applying Ito's lemma to $v_t = \sigma_t^2$ yields:

$$dv_t = 2\kappa\sigma_t\theta_t dt - 2\kappa\sigma_t^2 dt + 2\nu\sigma_t dW_t^2 + \nu^2 dt$$

(iv) Integrating the above equation yields:

$$\begin{aligned} \sigma_T^2 - \sigma_t^2 &= \nu^2(T-t) + 2\kappa \int_t^T \sigma_u\theta_u du - 2\kappa \int_t^T \sigma_u^2 du + 2\nu \int_t^T \sigma_u dW_u^2 \\ \iff \int_t^T \sigma_u dW_u^2 &= \frac{1}{2\nu}\left(\sigma_T^2 - \sigma_t^2 - \nu^2(T-t) - 2\kappa \int_t^T \sigma_u\theta_u du + 2\kappa \int_t^T \sigma_u^2 du \right) \end{aligned}$$

$$
\begin{aligned}
f_1(\phi) &= \mathbb{E}_t^{Q_1}[e^{i\phi x_T}] \\
&= \mathbb{E}_t^{Q}\left[\exp\left(-r(T-t) + (1+i\phi)x_T - x_t\right)\right] \qquad \text{(using (19))} \\
&= e^{i\phi(r(T-t)+x(t))}\mathbb{E}_t^{Q}\left[\exp\left\{(1+i\phi)\left(-\frac{1}{2}\int_t^T \sigma_u^2 du + \int_t^T \sigma_u dW_u^1\right)\right\}\right] \qquad \text{(using (9))} \\
&= e^{i\phi(r(T-t)+x(t))}\mathbb{E}_t^{Q}\left[\exp\left\{(1+i\phi)\left(-\frac{1}{2}\int_t^T \sigma_u^2 du + \rho\int_t^T \sigma_u dW_u^2 + \sqrt{1-\rho^2}\int_t^T \sigma_u dW_u\right)\right\}\right] \\
&= e^{i\phi(r(T-t)+x(t))}\mathbb{E}_t^{Q}\left[\exp\left\{\frac{1+i\phi}{2}\left((1+i\phi)(1-\rho^2)-1\right)\int_t^T \sigma_u^2 du + (1+i\phi)\rho\int_t^T \sigma_u dW_u^2\right\}\right] \\
&= \exp\left\{i\phi(r(T-t)+x(t)) - (1+i\phi)\frac{\rho}{2\nu}\left(\sigma_t^2 + \nu^2(T-t)\right)\right\} \times \Big(\\
&\qquad \mathbb{E}_t^{Q}\left[\exp\left\{\frac{1+i\phi}{2}\left((1+i\phi)(1-\rho^2) - 1 + 2\frac{\kappa\rho}{\nu}\right)\int_t^T \sigma_u^2 du\right\}\right] + \\
&\qquad \mathbb{E}_t^{Q}\left[\exp\left\{(1+i\phi)\frac{\rho}{2\nu}\sigma_T^2 - (1+i\phi)\frac{\kappa\rho}{\nu}\int_t^T \sigma_u\theta_u du\right\}\right]\Big) \\
&= \exp\left\{i\phi(x(t)+r(T-t)) - (1+i\phi)\frac{\rho}{2\nu}\left(\sigma_t^2 + \nu^2(T-t)\right)\right\}\times \\
&\qquad \mathbb{E}_t^{Q}\left[\exp\left\{s_1\int_t^T \sigma_u^2 du + s_2\int_t^T \sigma_u\theta_u du + s_3\sigma_T^2\right\}\right]
\end{aligned}
$$

where,

$$
s_1 = \frac{1+i\phi}{2}\left((1+i\phi)(1-\rho^2) - 1 + 2\frac{\kappa\rho}{\nu}\right) \tag{20}
$$

$$
s_2 = -(1+i\phi)\frac{\kappa\rho}{\nu} \tag{21}
$$

$$
s_3 = (1+i\phi)\frac{\rho}{2\nu} \tag{22}
$$

There remains to compute the expectation

$$
\begin{aligned}
y(t,\sigma,\theta) &= \mathbb{E}_t^Q\left[\exp\left\{s_1\int_t^T \sigma_u^2 du + s_2\int_t^T \sigma_u\theta_u du + s_3\sigma_T^2\right\}\right] \\
&:= \mathbb{E}_t^Q\left[\exp\left\{\int_t^T C(u,\mathcal{V}_u)du\right\}g(\mathcal{V}_T)\right]
\end{aligned}
$$

with state vector $\mathcal{V} = (\sigma,\theta)$ and $g(\mathcal{V}_T) = e^{s_3\sigma_T^2}$. Thus, the Feynman-Kac theorem may be used, notably that $\mathcal{V}$ is an Ito process with drift $b(\mathcal{V}) = (\kappa(\theta-\sigma),\lambda)$ and diffusion matrix $\Sigma = diag(\nu,\eta)$, as implied by equations (2) and (3). The generator for the process $\mathcal{V}$ is

$$
\begin{aligned}
\mathcal{L}u(\mathcal{V}) &= b(\mathcal{V})\cdot\nabla u + \frac{1}{2}Tr(A\nabla^2 u) \\
&= \kappa(\theta-\sigma)u_\sigma + \lambda u_\theta + \frac{1}{2}\sigma^2 u_{\sigma\sigma} + \frac{1}{2}\eta^2 u_{\theta\theta}
\end{aligned}
$$

where $A = \Sigma\Sigma^\top$. The Feynman-Kac theorem yields the PDE:

$$
\partial_t y + \mathcal{L}y + c(t,\mathcal{V})y = 0, \quad y(T,\mathcal{V}) = g(\mathcal{V}) \tag{23}
$$

$$
\iff \partial_t y + \kappa(\theta-\sigma)y_\sigma + \lambda y_\theta + \frac{1}{2}\sigma^2 y_{\sigma\sigma} + \frac{1}{2}\eta^2 y_{\theta\theta} + (s_1\sigma^2 + s_2\sigma\theta)y = 0, \tag{24}
$$

$$
y(T,\sigma,\theta) = e^{s_3\sigma_T^2}
$$

Emphasize the dependence of the $f_j's$ on τ, x_t, σ_t, and θ_t

3.2 Computation of the European Price

We describe an approach to compute the European price U using the "exact" formula presented in [**lin2024analytically**], (and attempt to inspect the bias it introduces).

The price of U depends on quantities $P_j(\tau,K)$ and $f_j(\ln S_t,\ \sigma_t,\ \theta_t,,\tau)$, $j = 1,2$.

The f_j's depend on the quantities $D(\tau,\phi)$, $E(\tau,\phi)$, $F(\tau,\phi)$, $G(\tau,\phi)$, $H(\tau,\phi)$, and $C(\tau,\phi)$

for their evaluation. The latter parameters are defined as the solutions on the initial value problem (25).

More precisely:

$$U(\tau; S_t, K) = S_t\, P_1(\tau; K) - K e^{-r\tau}\, P_2(\tau; K)$$

where for $j = 1, 2$:

$$P_j(\tau; K) = \tfrac{1}{2} + \frac{1}{\pi}\int_0^\infty Re\left[\frac{e^{-i\phi \ln K}}{i\phi}\, f_j(\tau;\phi)\right] d\phi$$

$$f_j(\tau;\phi) = \exp\big(C + D\, v_t^2 + E\, v_t\theta_t + F\, \theta_t^2 + G\, \theta_t + H\, v_t + i\phi\, x\big)$$

$$\begin{aligned}
\frac{\partial D}{\partial \tau}(\tau;\phi) &= \big(u_j\, i\phi - \tfrac{1}{2}\phi^2\big) + 2\nu^2 D^2(\tau;\phi) - 2A(\phi)\, D(\tau;\phi) + \tfrac{1}{2}\eta^2 E^2(\tau;\phi),\\
\frac{\partial E}{\partial \tau}(\tau;\phi) &= 2\kappa\, D(\tau;\phi) - A(\phi)\, E(\tau;\phi) + 2\nu^2 D(\tau;\phi)E(\tau;\phi) + 2\eta^2 E(\tau;\phi)F(\tau;\phi),\\
\frac{\partial F}{\partial \tau}(\tau;\phi) &= \kappa\, E(\tau;\phi) + \tfrac{1}{2}\nu^2 E^2(\tau;\phi) + 2\eta^2 F^2(\tau;\phi),\\
\frac{\partial G}{\partial \tau}(\tau;\phi) &= \kappa\, H(\tau;\phi) + 2\lambda\, F(\tau;\phi) + \nu^2 E(\tau;\phi)H(\tau;\phi) + 2\eta^2 F(\tau;\phi)G(\tau;\phi),\\
\frac{\partial H}{\partial \tau}(\tau;\phi) &= -A(\phi)\, H(\tau;\phi) + \lambda\, E(\tau;\phi) + 2\nu^2 D(\tau;\phi)H(\tau;\phi) + \eta^2 E(\tau;\phi)G(\tau;\phi),\\
\frac{\partial C}{\partial \tau}(\tau;\phi) &= r\, i\phi + \lambda\, G(\tau;\phi) + \tfrac{1}{2}\nu^2 H^2(\tau;\phi) + \nu^2 D(\tau;\phi) + \tfrac{1}{2}\eta^2 G^2(\tau;\phi) + \eta^2 F(\tau;\phi)
\end{aligned} \tag{25}$$

with $u_1 = \frac{1}{2}$, $u_2 = -\frac{1}{2}$, $b_1 = 1$, $b_2 = 0$, $A(\phi) = \kappa - \nu\rho\,(b_j + i\phi)$, $x = \ln S_t$ and the initial conditions $C(0;\phi) = D(0;\phi) = E(0;\phi) = F(0;\phi) = G(0;\phi) = H(0;\phi) = 0$. Here, $\tau = T - t$ is the remaining time until maturity of the option and ϕ is a dummy (frequency) parameter.

In practice, the European price $U(S_t, \sigma_t, \theta_t, t)$ may be estimated using a Monte Carlo simulation of the price process as described in section 2.1. Notice that also U may be

computed numerically by integrating the system (25) as follows:

(i) Write the dynamical system in [**lin2024analytically**] involving D, E, F, G, H, and C as

$$\frac{dX}{d\tau} = f_\Theta(\tau, \phi, X)$$

where $X^\top = [D\ E\ F\ G\ H\ C]$ and $\Theta = (\kappa, \nu, \lambda, \eta, \rho)$.

(ii) For given values of Θ the dynamical system may be integrated numerically to estimate values of D, E, F, G, H, and C on a $(\tau, \phi)-$grid.

(iii) The estimated values of D, E, F, G, H, and C can be used to compute the f_j's then integrate the P_j's, which in turn yield the value the option.

Remark 3.1

The European price $U(S_t, \sigma_t, \theta_t, t)$ will need to be evaluated along the paths generated to evaluate the American option. The price may be estimated using a Monte Carlo simulation of the price process using the same paths to value the American option if the European option were used as a control variate (see [**lsmc**] where this approach is applied to the European counterpart of the American-Bermuda-Asian option which is an option on the average stock price and can only be exercised at the final option maturity date.)

3.3 Pricing Path Dependent Options

The American price will be computed using a variant of the LSM algorithm as described in section 6.1.1 with variance reduction using the European price (25) as control variate following Rasmussen's [**rasmussen2005control**] approach. We adopt embedding the price of the corresponding European option into the pricing algorithm as described in section 3.3.1,

where the control parameter is computed on a per path basis (which yields tighter estimates!). The notation has been modified for consistency with [**lsmc**] and [**chavez2022monte**].

3.3.1 LSM-CV Algorithm

Suppose that we are given the stock price matrix $S^i_{t_j}$ and the option's intrinsic value matrix I^i_j, (i.e. the immediate exercise price at node $S^i_{t_j}$).

1. Define the vectors of length N at time point t_j for $j = N, N-1, \ldots, 1$:

   - $E_j = V^E_j(S_{t_j})$, the vector with the ith entry is $V^E_j(S^i_{t_j})$ as given by the closed formula for the value of a European option whose term is given by $t_N - t_j$ and spot by $S^i_{t_j}$. The strike and style (call or put) of this European option is equal to that of the American option under consideration;
   - CF_j denote the vector with the ith entry equal to $e^{-r(\bar{\tau}^i_{j+1}-t_j)} I^i_{\bar{\tau}^i_{j+1}}$, the discounted eventual exercise value of path i. CF stands for cash flow, $I^i_{\bar{\tau}^i_{j+1}}$ is the exercise price, and $\bar{\tau}^i_{j+1}$ is the exercise time (or the stopping time) along the ith path;
   - TV_j indicates the vector with the ith entry equal to $\tilde{V}^{\theta^i}_j(S^i_{t_j})$, where $\tilde{V}^{\theta}_j(S_{t_j})$ is the modeled continuation value (or test value) used to decide whether the option should be exercised at node $S^i_{t_j}$.

     Here, the computation of $\tilde{V}^{\theta}_j(S_{t_j})$ involves (i) the estimator $\tilde{V}_j(S^i_{t_j})$ obtained by regressing the cash flows $CF^i_j = e^{-r(\bar{\tau}^i_{j+1}-t_j)} I^i_{\bar{\tau}^i_{j+1}}$ on a set of basis functions, (ii) the control variate and the θ's as suggested by Rasmussen's approach in [**rasmussen2005control**];
   - CV_j indicate the vector with the ith entry equal to $\tilde{V}^E_j(S^i_{t_j})$, where $\tilde{V}^E_j(S_{t_j})$ is the

modeled price used to evaluate an European option written at the exercise time $\bar{\tau}^i_{j+1}$ at node $S^i_{t_j}$.

Here, CV stands for control variate. $\tilde{V}^E_j(S_{t_j})$ is estimated by regressing the values $e^{-r(\bar{\tau}^i_{j+1}-t_j)}E^i_{\bar{\tau}^i_{j+1}}$ on the same set of basis functions;

Note that the dependence of these vectors on the time index j may be suppressed as they can be overwritten at each step.

2. At maturity t_N initialize IV and CV as the exercise price (or payoff) for each option, based on the simulated paths of S.

3. For $j = N-1, \ldots, 1$:

   (i) Apply the updates $IV_j = e^{-r\Delta t_{j+1}} IV_{j+1}$ and $CV_j = e^{-r\Delta t_{j+1}} CV$

   (ii) Compute the estimators: $\tilde{V}_j(S_{t_j})$, $\tilde{V}^E_j(S_{t_j})$

   (iii) Compute the control parameters $\hat{\theta}^i_j$:

   - compute the auxiliary estimators $\widehat{CV}_j^{\text{aux}_1}(S_{t_j})$ and $\widehat{CV}_j^{\text{aux}_2}(S_{t_j})$ by regressing the values $E^i_j CV^i_j$, and the values $(CV^i_j)^2$ on the same set of basis functions.

   - $$\hat{\theta}^i_j = -\frac{\widehat{CV}_j^{\text{aux}_1}(S^i_{t_j}) - \tilde{V}_j(S^i_{t_j})\tilde{V}^E_j(S^i_{t_j})}{\widehat{CV}_j^{\text{aux}_2}(S^i_{t_j}) - \left[\tilde{V}^E_j(S^i_{t_j})\right]^2}$$

   - Compute the test value TV:

$$TV^i_j = \tilde{V}^{\theta^i}_j(S^i_{t_j}) := \tilde{V}_j(S^i_{t_j}) + \hat{\theta}^i(\tilde{V}^E_j(S^i_{t_j}) - E^i_j)$$

- If the immediate exercise value $I_j^i > \max(E_j^i, TV_j^i)$, then apply the updates:

$$CF_j^i = I^i \text{ and } CV_j^i = E_j^i$$

4. The option value determined by the LSM combined with the control variate suggested by Rasmussen [**rasmussen2005control**] is

$$\max\left(I_0,\ \frac{e^{-rt_1}}{N}\sum_{i=1}^{N} CF_1^i\right)$$

5. An improved estimator is reported in [**west2013american**] following the example of [**rasmussen2002improving**]:

$$\max\left(I_0,\ \frac{e^{-rt_1}}{N}\sum_{i=1}^{N} CF_1^i + \hat{\theta}\left(\frac{e^{-rt_1}}{N}\sum_{i=1}^{N} CV_1^i - E_1^i\right)\right)$$

where $\hat{\theta}$ is the regression coefficient obtained by regressing CF_1 on CV_1

4 Numerical Experiments

4.1 Validation of the implementation on edge cases

Notice that for the particular choice of the parameters $\kappa = \nu = \lambda = \eta = \rho = 0$, the LLH model (1), (2), (3) reduces to the benchmark Black-Scholes (BS) model. Furthermore, if $\lambda = \eta = \rho = 0$ the LLH model reduces to the Stein-Stein model introduced in [**stein1991stock**].

We validate our implementation of the European price formula (25) by comparing our output under these particular values of the parameters with the output of the closed formula of the resulting corresponding benchmark model.

4.1.1 The Black-Scholes Limit: $\kappa = \nu = \lambda = \eta = \rho = 0$

We generate $N = 10000$ paths from the LLH model and we use the Kolmogorov-Smirnov test to determine whether the price slice at the end of the simulation horizon $T = 1.0$ follows a lognormal distribution. The test statistic is $KS = 0.0051$ and the $p-$value $= 0.9547$ which suggests that lognormalilty hypothesis cannot be rejected.

For spot price $S = 100.0$, strike $K = 90.0$, 1 year maturity, and volatility $\sigma = 0.2$, the output of our implementation of the LLH European price (25) coincides with that of the BS price at 16.6994484084.

4.1.2 Stein-Stein Limit: $\lambda = \eta = \rho = 0$

For the sake of comparison, we set the values of the remaining model parameters using Table 1 in [**stein1991stock**]: $\kappa = 4$, $\nu = 0.1$, $r = 0.0953$.

For spot price $S = 100.0$, strike $K = 90.0$, 1 month maturity, volatility parameters $\sigma_0 = 0.2$ and $\theta_0 = 0.2$, the LLH price of a the European call price is 10.7668609302. The

Stein-Stein price obtained in [**stein1991stock**] is 10.7700000000.

4.2 Illustration of the LLH model

r=0.3943, kappa=4.9394, nu=0.4, sigma0=0.2924, theta0=0.1319, lam=0.3115, eta=0.4112, rho=0.1691 from the Lin-Lin-He paper Table 2

4.3 Illustration of the formula of the European price

We illustrate the computation of European options using the algorithm outlined in section 3.1 of the report (the "exact" LLH formula of the European price). Furthermore, we perform a Monte Carlo simulation of $1,000,000$ paths to estimate the price of vanilla european options under the LLH model and we compare the results with the results of the LLH price formula.

(i) Consider a 1 month horizon ($T = 0.083$ yr) with 22 trading days.

(ii) Consider a 1 year horizon with 252 trading days.

Remarks

- Recall the call-put parity: $P - C = Ke^{-t\tau} - S$ which implies that $P_E^{llh} - P_E^{MC} = C_E^{llh} - C_E^{MC}$
- Notice the bias in the LLH price that accentuates with longer maturity. The implementation of the "exact" LLH price seems to be consistently undervaluing the European price. The bias is due to the errors that result from the estimation of the ODE system and the integrals involved in the LLH formula. Further investigation is required.

5 Discussion

5.1 The Black-Scholes Limit

$$D_\tau(\tau;\phi) = u_j\, i\phi \;-\; \tfrac{1}{2}\,\phi^2,$$

$$C_\tau(\tau;\phi) = r\, i\phi,$$

$$E_\tau(\tau;\phi) = 0, \qquad F_\tau(\tau;\phi) = 0,$$

$$G_\tau(\tau;\phi) = 0, \qquad H_\tau(\tau;\phi) = 0$$

$$C(0) = D(0) = E(0) = F(0) = G(0) = H(0) = 0.$$

$$D(\tau;\phi) = \big(u_j\, i\phi - \tfrac{1}{2}\,\phi^2\big)\tau, \quad C(\tau;\phi) = r\, i\phi\,\tau, \quad E(\tau;\phi) = F(\tau;\phi) = G(\tau;\phi) = H(\tau;\phi) = 0.$$

$$f_j(\tau;\phi) = \exp\Big(i\phi\, x + r\, i\phi\,\tau + \big(u_j\, i\phi - \tfrac{1}{2}\,\phi^2\big)\,\sigma_0^2\,\tau\Big), \qquad j = 1, 2.$$

$$P_j(\tau;K) = \tfrac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\left[\frac{e^{-i\phi\,\ln K}}{i\phi}\, f_j(\tau;\phi)\right] d\phi, \qquad j = 1, 2.$$

$$P_1 = \mathcal{N}(d_1), \qquad P_2 = \mathcal{N}(d_2), \quad \text{where} \quad d_1 = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma_0^2)\tau}{\sigma_0\sqrt{\tau}}, \quad d_2 = d_1 - \sigma_0\sqrt{\tau}.$$

6 Annex

6.1 Valuing American Options by the Least-Squares Monte Carlo Method

In this section, we review the main points of the LSM valuation approach described in [**lsmc**]. First, we set the notation. Assume a finite time horizon $[0, T]$, and let Ω be the set of all possible realizations of the stochastic economy between 0 and T. Denote by $\mathcal{F}_t$ the sigma algebra of distinguishable events at time t, and by $F = \{\mathcal{F}_t,\, 0 \leq t \leq T\}$ the filtration generated by the price processes of the securities in the economy. Assuming no-arbitrage, let Q denote the martingale measure for this economy,

Denote by $C(\omega,\, s;\, t,\, T)$ the path of cash flows generated by an option, conditional on the option not being exercised prior or at t, and on the option holder following optimal stopping strategy for all s, $t < s \leq T$. Focus on (Bermudan) options that can be exercised at the K discrete times $0 \leq t_1 \leq t_2 \leq \cdots \leq t_K = T$.

6.1.1 The LSM Algorithm in a Nutshell

Standard results imply that the value of an American option equals the maximized value of the discounted cash flows from the option, where the maximum is taken over all stopping times w.r.t. the filtration F. The no-arbitrage condition implies that the value $F(\omega;\, t_k)$ of the option assuming it cannot be exercised until after t_k is:

$$F(\omega;\, t_k) = \mathbb{E}^Q \left[\sum_{j=k+1}^{K} \exp\left(-\int_{t_k}^{t_j} r(\omega, s) ds \right) C(\omega,\, t_j;\, t_k,\, T) \middle| \mathcal{F}_{t_k} \right] \tag{26}$$

where $r(\omega, t)$ is the (possibly stochastic) riskless discount rate.

Notice that at time t_k, the cash flow from immediate exercise is known, while the cash

flows from continuation are not. With the representation in (26), the problem of optimal exercise reduces to comparing the immediate exercise value with this conditional expectation, and then exercising as soon as the immediate exercise value is positive and greater than or equal to the conditional expectation.

The objective of the LSM algorithm is to provide a pathwise approximation to the optimal stopping rule that maximizes the value of the American option. It uses least squares to approximate the conditional expectation function in (26) at t_{K-1}, $t_{K-2}, \ldots,$ t_1.

LSM algorithm:

1. At time t_{K-1}, assume $F(\omega;\, t_{K-1})$ can be represented as a linear combination of a countable set of basis functions, and approximate it using the first M basis functions. The estimate $F_M(\omega; t_{K-1})$ is computed by regressing the discounted values $C(\omega,\, s;\, t_{K-1},\, T)$ onto the basis functions for the paths where the option is in the money at time t_{K-1} to obtain $\widehat{F}_M(\omega; t_{K-1})$. Once the $\widehat{F}_M(\omega; t_{K-1})$ has been computed, we can determine whether early exercise at time t_{K-1}, is optimal for an in-the-money path ω by comparing the immediate exercise value with $\widehat{F}_M(\omega; t_{K-1})$, and repeating for each in-the-money path.

2. Once the exercise decision is identified, the option cash flow paths $C(\omega, s; t_{K-2}, T)$ can then be approximated. The recursion proceeds by rolling back to time t_{K-2}, and repeating the procedure until the exercise decisions at each exercise time along each path have been determined.

3. The American option is then valued by starting at time zero, moving forward along each path until the first stopping time occurs, discounting the resulting cash flow from exercise back to time zero, and then taking the average over all paths ω.

Remark 6.1

1. The LSM algorithm can be used to approximate the value of continuously exercisable American options by taking K in (26) to be sufficiently large.

2. If the value of the asset underlying the option is not Markovian, both the current and past realizations of the state variables can be included in the basis functions and the regressions.

3. Since the values of the basis functions are independently and identically distributed across paths, weak assumptions about the existence of moments allow us to show that the fitted value $\widehat{F}_M(\omega; t_{K-1})$ converges in mean square and in probability to $F_M(\omega; t_{K-1})$ as the number N of (in-the-money) paths in the simulation goes to infinity. Furthermore, it can be shown that $\widehat{F}_M(\omega; t_{K-1})$ is the best linear unbiased estimator of $F_M(\omega; t_{K-1})$ based on a mean-squared metric.

6.2 Variance Reduction of the LSM Estimator

6.2.1 Rasmussen's Variance Reduction Approach